University of Applied Sciences

HOCHSCHULE EMDEN·LEER

Machine Learning

Prof. Dr. E. Wings

Agenda

1. Introduction
2. Neural Networks & Co.
3. Data Science
4. Introduction to Data Mining
5. Examples
6. Data Science

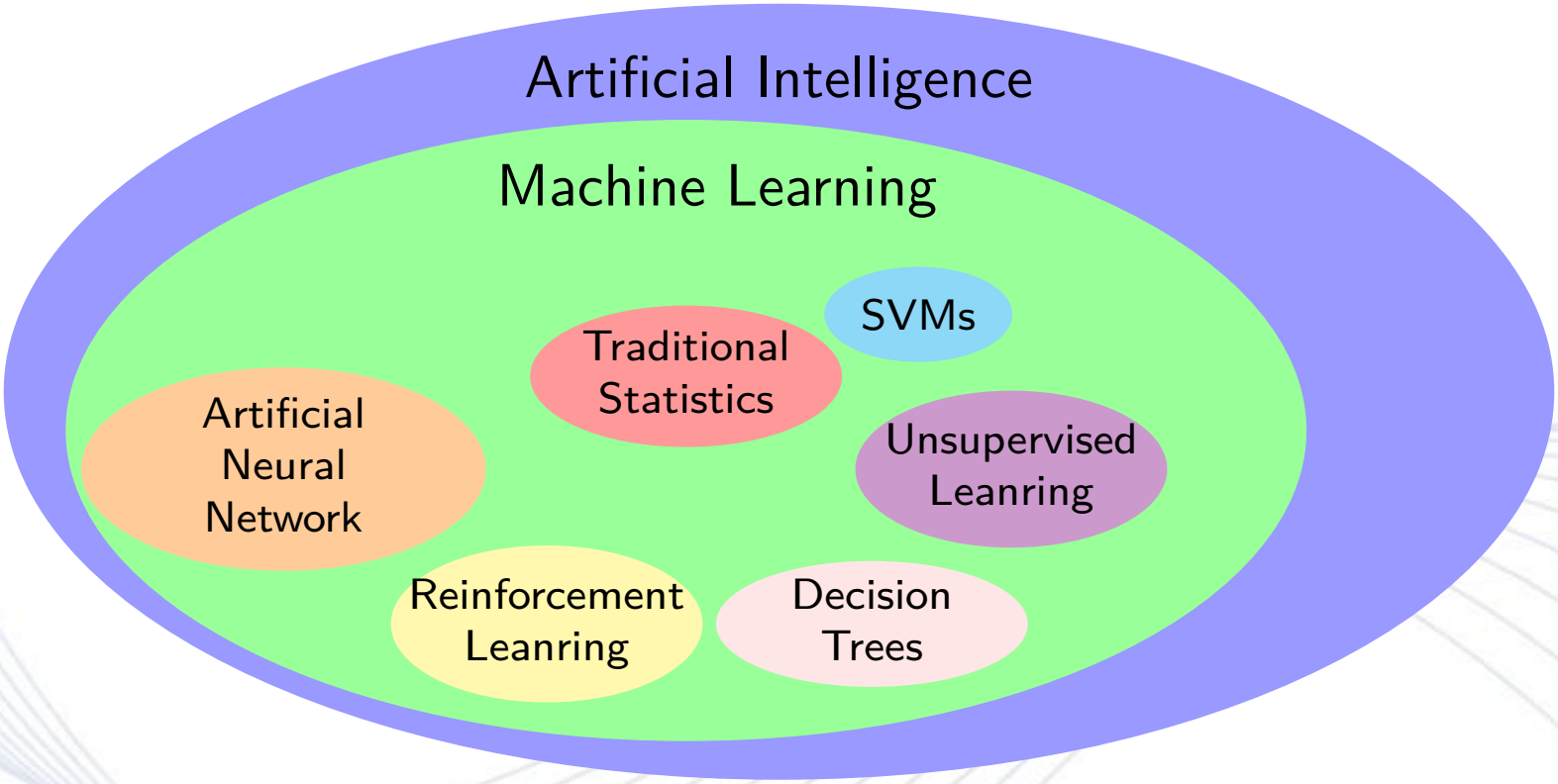
Introduction

Why Machine Learning?

- The logic required to make a decision is specific to a single domain and task. Changing the task even slightly might require a rewrite of the whole system.
- Designing rules requires a deep understanding of how a decision should be made by a human expert.

Machine learning algorithms automate decision-making processes by generalizing from known examples.

Fields of Data Mining



Artificial Intelligence

- Artificial intelligence is a broad field, but more modern.
- Although some ideas were presented in ancient times, the field has been documented as being born at a workshop at Dartmouth College in 1956.
- Artificial intelligence is presented in contact with the intelligence displayed by humans and other animals and the domain may include not just techniques that use broad amounts of data, but also others like agent-based modelling which have been used in aviation.

Machine Learning

- Machine Learning is part of the field of artificial intelligence.
- Machine learning is a set of techniques and algorithms that use large quantities of data to achieve certain AI goals.
- Machine learning focuses on the development of computer programs that can access data and use it learn for themselves.

Machine Learning

In machine learning, tasks are generally classified into broad categories. These categories are based on how learning is received or how feedback on the learning is given to the system developed:

- Supervised Learning
- Unsupervised Learning
- Reinforcement or Semi-Supervised Machine Learning

Supervised Learning

Supervised learning trains algorithms based on example input and output data that is labeled by humans.

The computer is provided with example inputs that are labeled with their desired outputs. The purpose of this method is for the algorithm to be able to "learn" by comparing its actual output with the "taugh" outputs to find errors, and modify the model accordingly.

Supervised learning therefore uses patterns to predict label values on additional unlabeled data.

Supervised Learning

There are two major types of supervised machine learning problems:

- classification
- regression

Classification

- binary classification, e.g. yes/no, spam/not spam, positive/negative class
- multiclass classification, e.g. different languages

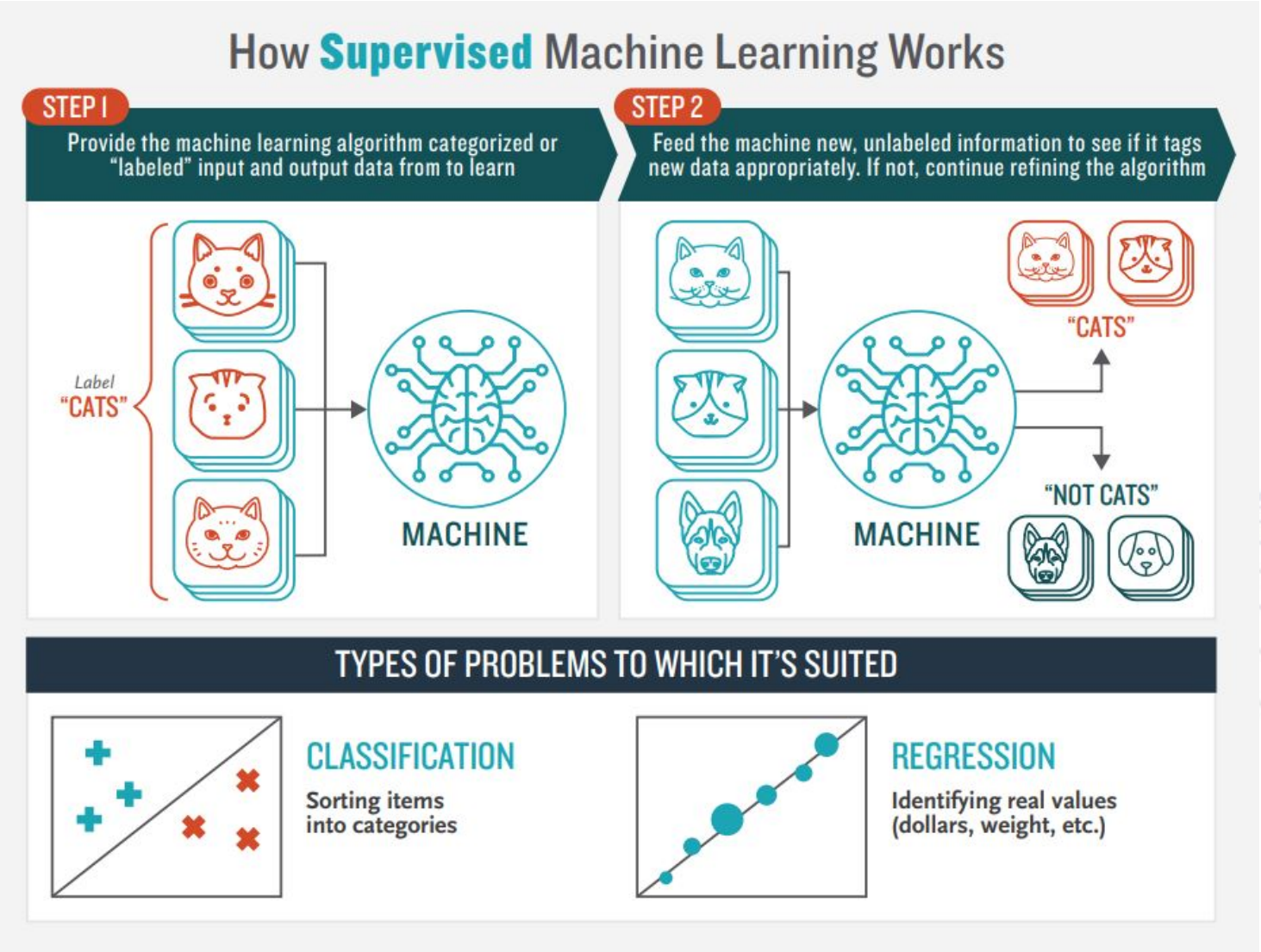
Regression

The goal is to predict a continuous number, or a floating-point number

Supervised Learning - Algorithms

- Support Vector Machines
- Linear regression
- Logistic regression
- Naive Bayes classification
- Linear discriminant analysis
- Decision trees
- k-Nearest neighbor algorithm
- Neural Networks (Multilayer perceptron)
- Similarity learning
- Random forest

Supervised Learning



Artificial Neural Network

- ANN removes the manual identification of features in data and, instead, relies on whatever training process it has in order to discover the useful patterns in the input examples.
- An ANN is a series of algorithms that endeavors to recognize underlying relationships in a set of data through a process that mimics the way the human brain operates.
- Artificial neural networks refer to systems of neurons, either organic or artificial in nature.
- Artificial neural networks can adapt to changing input; so the network generates the best possible result without needing to redesign the output criteria.

Unsupervised Leanring

- Unsupervised learning algorithms are used when the information used to train is neither classified nor labeled.
- Unsupervised learning studies how systems can infer a function to describe a hidden structure from unlabeled data.
- The system doesn't figure out the right output, but it explores the data and can draw inferences from datasets to describe hidden structures from unlabeled data.

Unsupervised Learning

Unsupervised learning provides the algorithm with no labeled data in order to allow it to find structure within its input data.

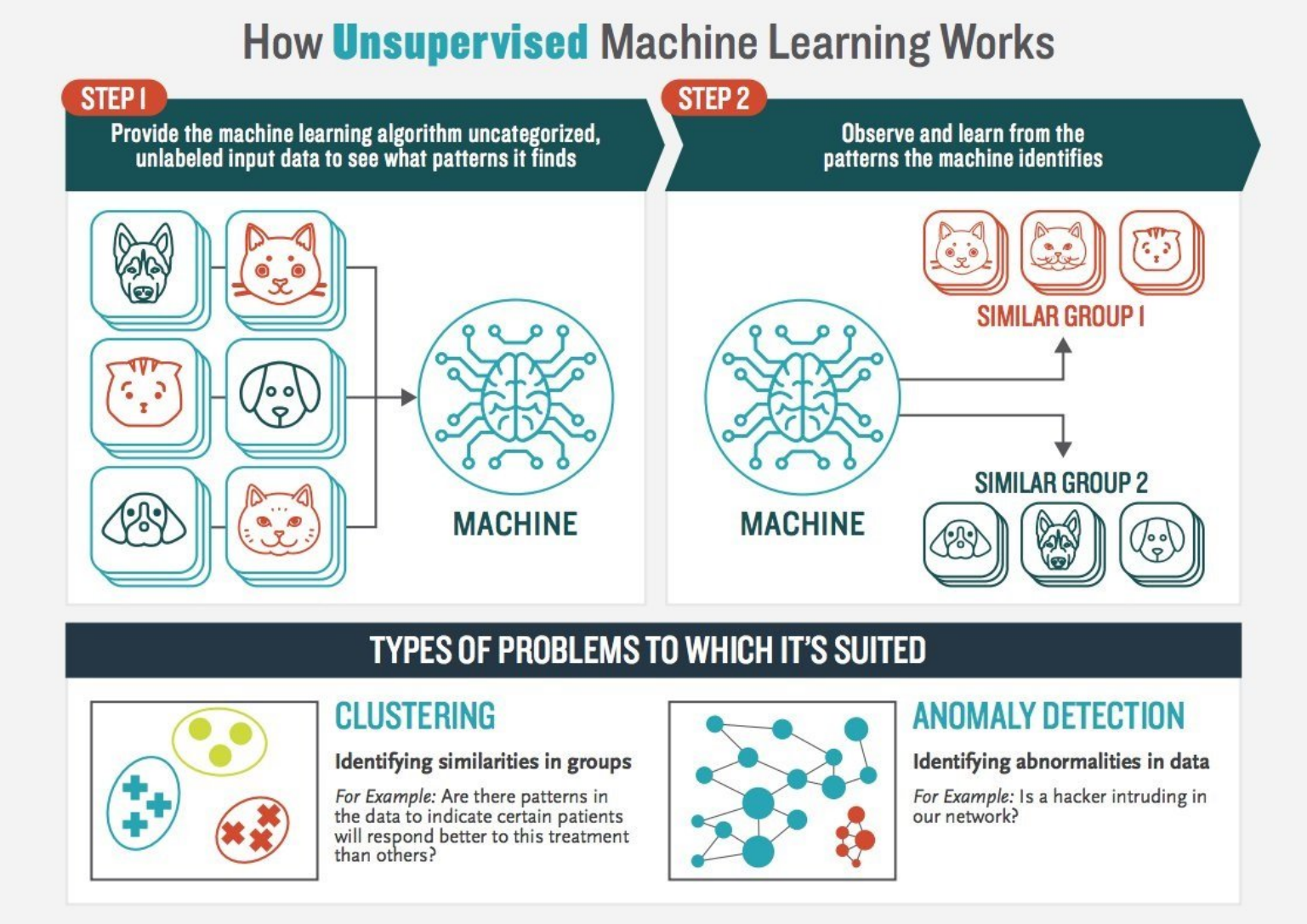
Data is unlabeled, so the learning algorithm is left to find commonalities among its input data. As unlabeled data are more abundant than labeled data, machine learning methods that facilitate unsupervised learning are particularly valuable.

The goal of unsupervised learning may be as straightforward as discovering hidden patterns within a dataset, but it may also have a goal of feature learning, which allows the computational machine to automatically discover the representations that are needed to classify raw data.

Unsupervised Learning - Algorithms

- K-means
- Apriori algorithm
- Principal Component Analysis
- Singular Value Decomposition
- Independent Component Analysis

Unsupervised Learning



Reinforcement Learning

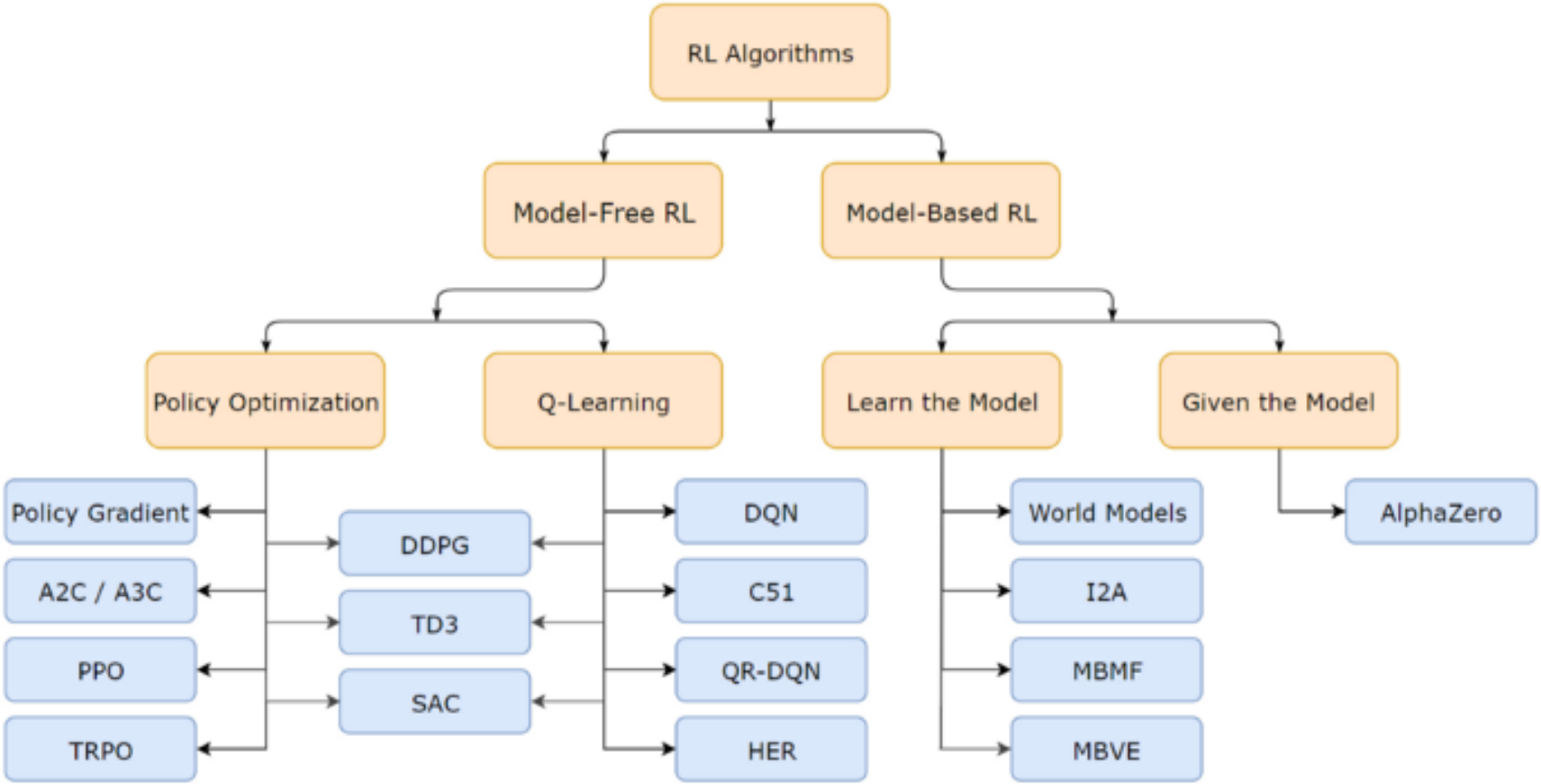
- Reinforcement machine learning algorithms is a learning method that interacts with its environment by producing actions and discovers errors or rewards.
- Trial and error search and delayed reward are the most relevant characteristics of reinforcement learning.
- Reinforcement Learning allows machines and software agents to automatically determine the ideal behavior within a specific context in order to maximize its performance.
- Simple reward feedback is required for the agent to learn which action is best; this is known as the reinforcement signal.

Reinforcement or Semi-Supervised Machine Learning

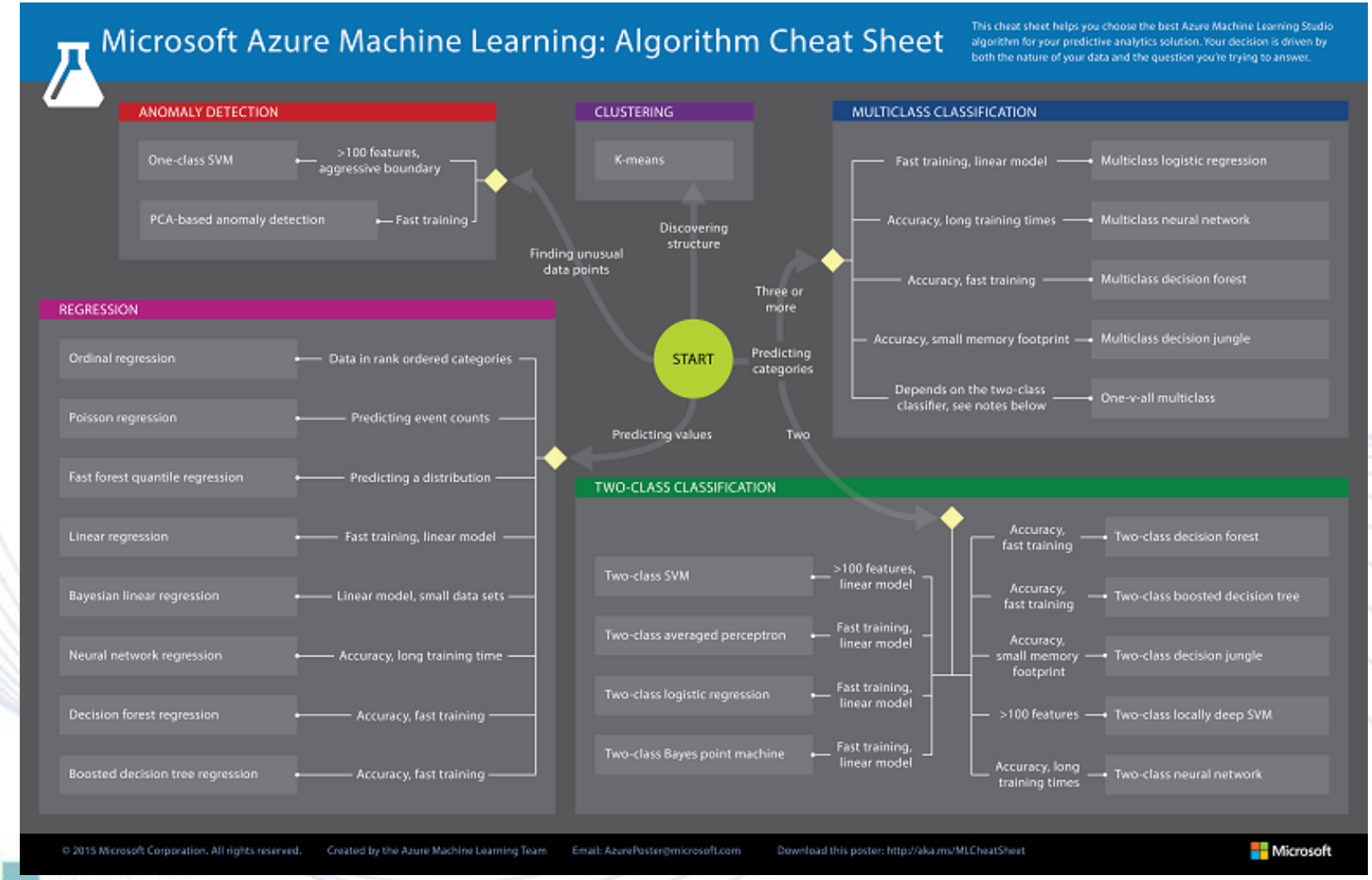
There are Problems that you've found a large amount of input data, but just some of are labeled.

These are termed as semi-supervised learning problems.

Reinforcement - Algorithms



Cheat Sheet


Microsoft Azure Machine Learning: Algorithm Cheat Sheet
This cheat sheet helps you choose the best Azure Machine Learning Studio algorithm for your predictive analytics solution. Your decision is driven by both the nature of your data and the question you're trying to answer.
ANOMALY DETECTION
One-class SVM
>100 features, aggressive boundary
PCA-based anomaly detection
Fast training
CLUSTERING
K-means
MULTICLASS CLASSIFICATION
Fast training, linear model
Multiclass logistic regression
Accuracy, long training times
Multiclass neural network
Accuracy, fast training
Multiclass decision forest
Accuracy, small memory footprint
Multiclass decision jungle
Depends on the two-class classifier, see notes below
One-v-all multiclass
Finding unusual data points
Discovering structure
Three or more
START
Predicting categories
Predicting values
Two
REGRESSION
Ordinal regression
Data in rank ordered categories
Poisson regression
Predicting event counts
Fast forest quantile regression
Predicting a distribution
Linear regression
Fast training, linear model
Bayesian linear regression
Linear model, small data sets
Neural network regression
Accuracy, long training time
Decision forest regression
Accuracy, fast training
Boosted decision tree regression
Accuracy, fast training
TWO-CLASS CLASSIFICATION
Two-class SVM
>100 features, linear model
Two-class averaged perceptron
Fast training, linear model
Two-class logistic regression
Fast training, linear model
Two-class Bayes point machine
Fast training, linear model
Accuracy, fast training
Two-class decision forest
Accuracy, fast training
Two-class boosted decision tree
Accuracy, small memory footprint
Two-class decision jungle
>100 features
Two-class locally deep SVM
Accuracy, long training times
Two-class neural network
© 2015 Microsoft Corporation. All rights reserved.
Created by the Azure Machine Learning Team
Email: AzurePoster@microsoft.com
Download this poster: http://aka.ms/MLCheatSheet
Microsoft

Traditional Statistics

- Initial developments of statistics is dating back to the 1600s.
- Statistics is a branch of mathematics dealing with the collection, analysis, interpretation, presentation, and organization of data.
- The field adds significant value to any research effort and one may argue that statistics could be the first and fundamental step of any other related technique to be developed.
- Many reports are based on statistics and provide considerable information to the aviation industry professionals.
- Statistics may exhibit some degree of forecasting, but only when the relationships between the variables are simple enough to be presented and directly readable by humans.

Statistics

- Foundation of most data mining technologies
  - Regression analysis
  - Standard distribution/deviation/variance
  - Cluster analysis
  - Confidence intervals
- Building blocks
- Significant role in today's data mining - but alone is not powerful enough

Query Reporting vs. Business Intelligence vs. Data Mining

Query Reporting	Business Intelligence	Data Mining
Extraction of data; detailed and/or summarized	Analysis, summaries, actual trends	Discovery of hidden patterns, information, predicting future trends
Information	Analysis	Insight knowledge and prediction
Who purchased the product in the last 2 quarters?	What is an average income of the buyers per quarter by district?	Which customers are likely to buy a similar product in the future and why?

Programming Languages I

- Java
  - Deeplearning4j
  - MALLET (MAchine Learning for LanguagE Toolkit)
  - Weka
- R
  - caret
  - randomForest
  - e1071
- C++
  - mlpack
  - Dlib
  - Shark

Programming Languages II

- Python
  - deep learning frameworks, e.g.
    - TensorFlow
    - PyTorch
    - Keras
  - Packages, e.g.
    - NumPy
    - SciPy
    - Matplotlib.

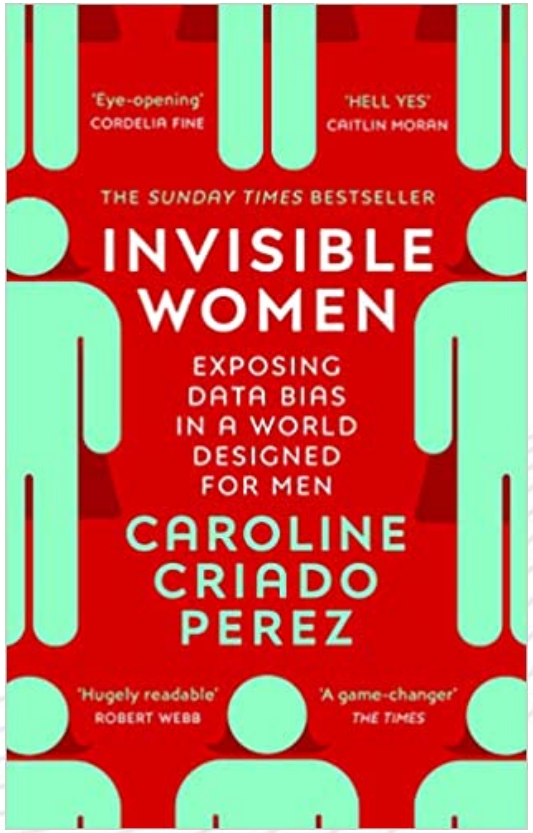
Human Biases

Although data and computational analysis may make us think that we are receiving objective information, this is not the case; being based on data does not mean that machine learning outputs are neutral.

Human bias plays a role in how data is collected, organized, and ultimately in the algorithms that determine how machine learning will interact with that data.

Human Biases

Caroline Criado Perez: Invisible Women: Exposing Data Bias in a World Designed for Men. Wiley, 2020



Human Biases

- Plan-confinuation bias
- Confirmation Bias
- Experience selective attention bias
- Implicit or unconscious Bias
- Hindsight bias
- Anchoring
- Overconfidence bias
- Gambler's fallacy
- Fundamental attribution error

Avoid Human Biases

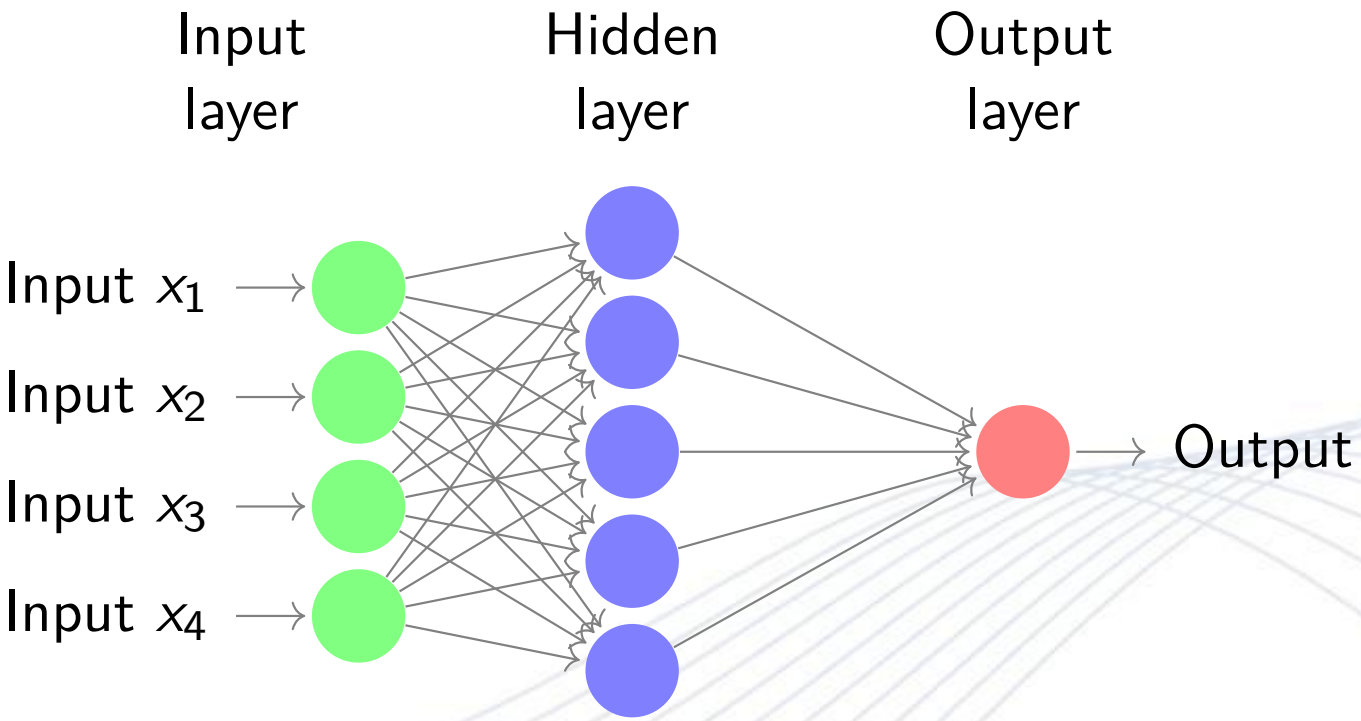
- Avoid Stress, lack of sleep:
  $\Rightarrow$ Eat drink, take a break,
- Checklists
- Plan

Neural Networks & Co.

Types of Neural Networks

- Standard Neural Network
- Deep Learning Network (DNN)
- Convolutional Neural Network (CNN)
- Recurrent Neural Network (RNN)
- Long-Short Term Memory (LSTM)

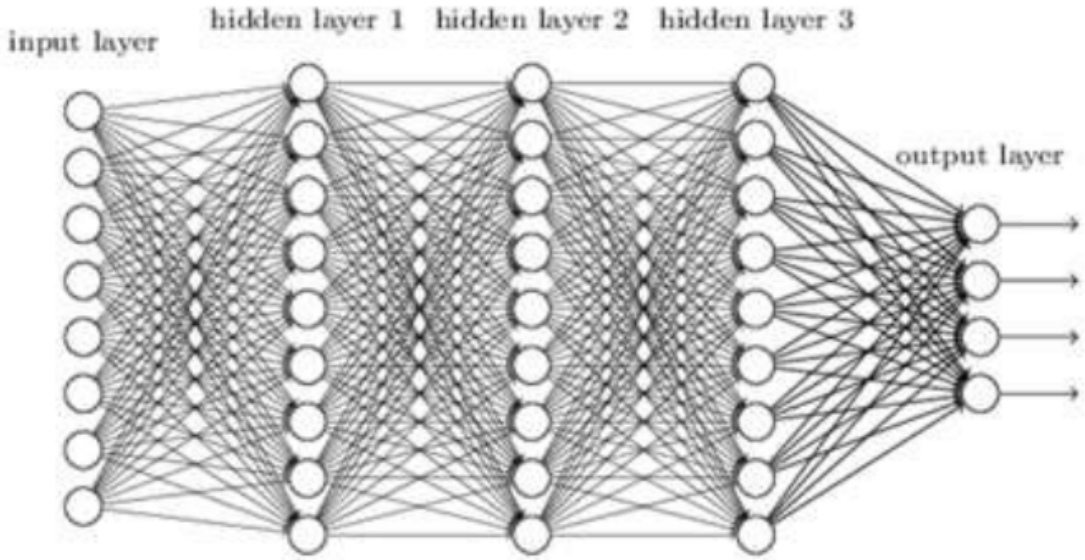
Standard Neural Network - Structure



Standard Neural Network - Applications

- Inverse kinematics for a robot with SCARA kinematics
- Approximation of functions
- Easy classification

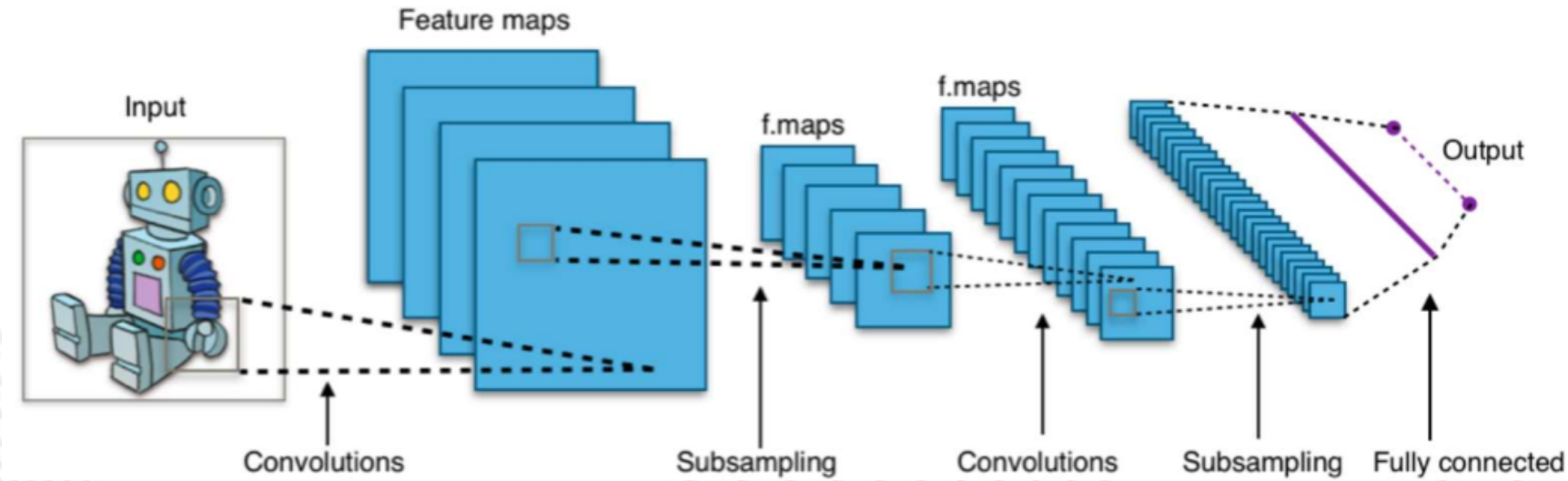
Deep Neural Network - Structure



Deep Neural Network - Applications

- Image recognition,
  e.g. Image data for diagnoses of diseases
- Language and sentiment analysis
- Risk minimization for financial transactions
- Handwriting recognition

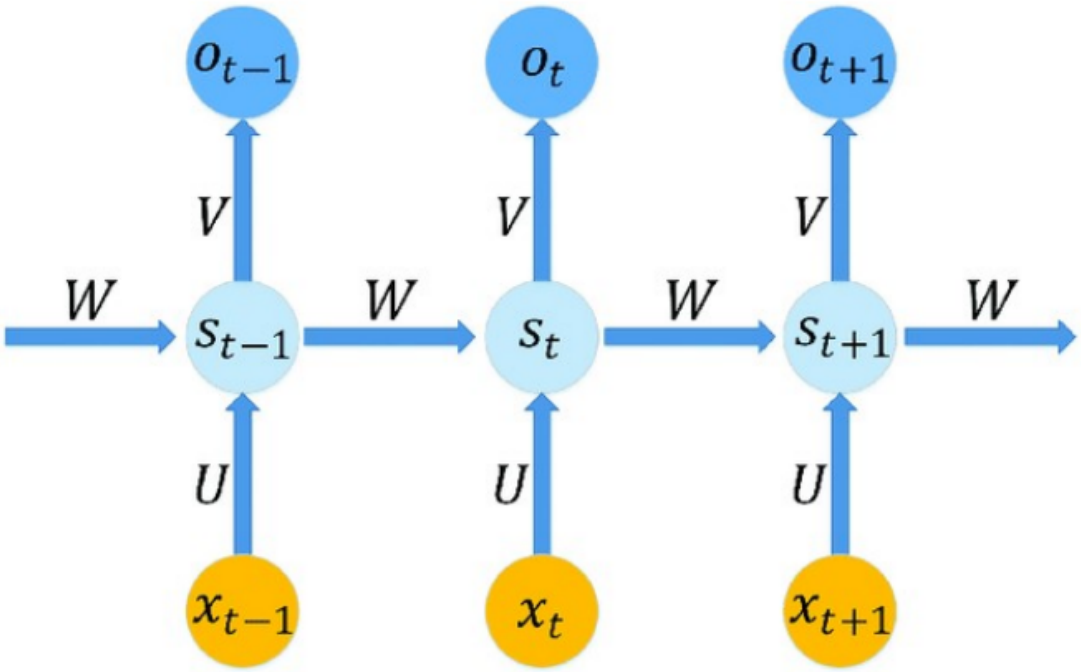
Convolutional Neural Network - Structure



Convolutional Neural Network - Applications

- Image recognition
- Parameter optimization for machine tools
- Search query recognition

Recurrent Neural Network - Structure



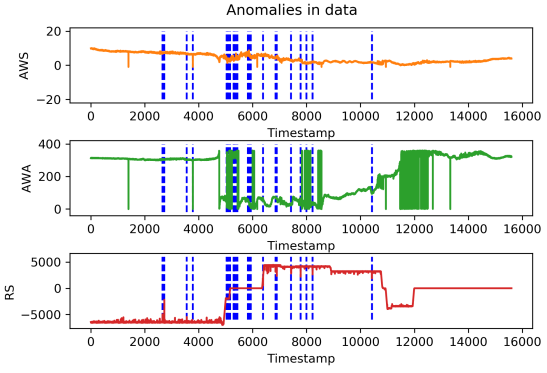
Recurrent Neural Network - Applications

Algorithm for sequential data like

- time series,
- voice,
- text,
- financial data,
- audio,
- video,
- weather

Long-Short Term Memory

LSTM - Networks is an extension for Recurrent Neural Networks, which essentially expand their storage.



Long-Short Term Memory - Applications

- Identification of brain tumors from MRT sectional images
- Time series data of sunspots
- Time series data of stock prices
- Time series data of buying behavior

Other DNN techniques and architectures

- Autoencoder
- Growing self-organizing map (GSOM)
- CapsNet (add structures called "capsules" to a CNN to reuse output from several of those capsules to form more stable representations for higher order capsules.)

Other DNN techniques and architectures

- Generative Adversarial Networks (a deep neural net architectures comprised of two nets, pitting one against the other)
- Image generation (e.g., https://thispersondoesnotexist.com/)
- Latent spaces (Utilize/train intermediate layers of deep network as feature layer for complementary (secondary) machine learning task)

Data Science

Knowing Your Task and Knowing Your Data

- What question(s) am I trying to answer? Do I think the data collected can answer that question?
- What is the best way to phrase my question(s) as a machine learning problem?
- Have I collected enough data to represent the problem I want to solve?
- What features of the data did I extract, and will these enable the right predictions?
- How will I measure success in my application?
- How will the machine learning solution interact with other parts of my research or business product?

Knowing Your Task and Knowing Your Data

Attention: Machine learning are only one part of a greater process to solve a particular problem

Data Science

- Scientific approach to handling heterogeneous and unstructured data
- Investigation of sub-areas to generate knowledge from raw data
- Subareas of data handling:
  - Recording,
  - storage,
  - organization,
  - evaluation processes e.g. KDD, CRISP-DM,
  - analysis,
  - visualization and
  - interpretation
- Use of machine learning and algorithms

$\Rightarrow$ Data Science is an all-encompassing approach to working with Big Data

Data Mining - Procedure models

- Knowledge Discovery in Databases (KDD)
  - Process model for discovery from data
  - Procedure for selecting and preparing data
  - Data mining describes analysis methods and is a sub-process step of KDD model
- CRoss-Industry Standard Process for Data Mining (CRISP-DM)
  - Iterative and agile process model for data mining
  - Standard process with six phases for analyzing data sets according to patterns, trends and relationships

$\Rightarrow$ Process models structure the evaluation of Big Data

Data Mining - Procedure models

KDD	**CRISP-DM**
General procedure for obtaining knowledge from Big Data	Focus on embedding in the corporate context Comparison of the results with defined expectations

- Structured evaluation of large amounts of data
- Data Mining as sub-process
- Identification of patterns within databases

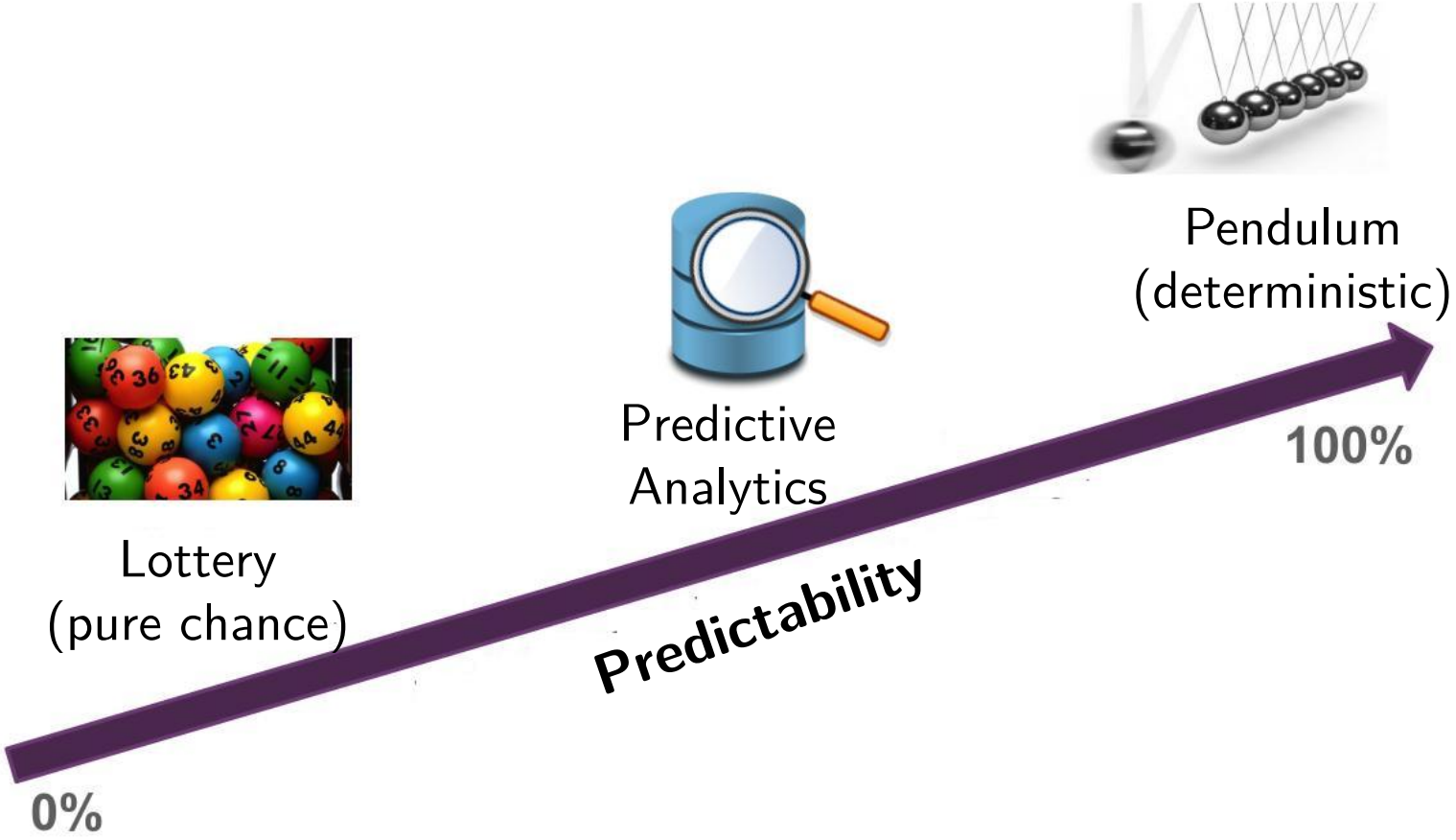
Introduction to Data Mining

Data Mining Applications

- Market analysis and management
  - Target marketing, customer relation management, market basket analysis, cross selling, market segmentation (credit card scoring, Personalization & Customer Profiling)
- Risk analysis and management
  - Forecasting, customer retention, improved underwriting, quality control, competitive analysis (banking and credit card scoring)
- Fraud detection and management
  - Fraud, waste and abuse including: illegal practices, waste, payment error, non-compliance, incorrect billing practices

Examples

Predictive Analytics



Example Wind Power Plants

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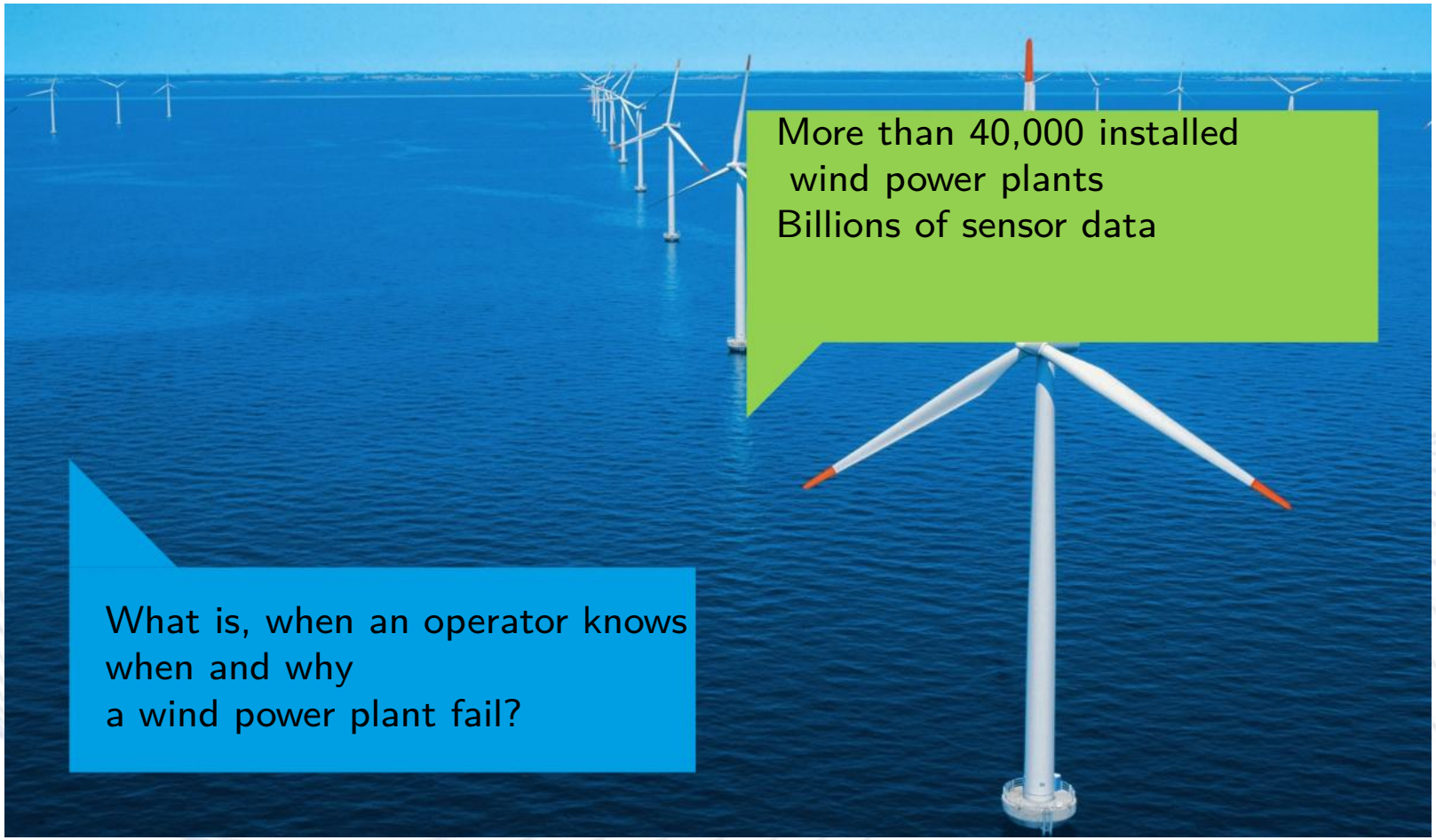

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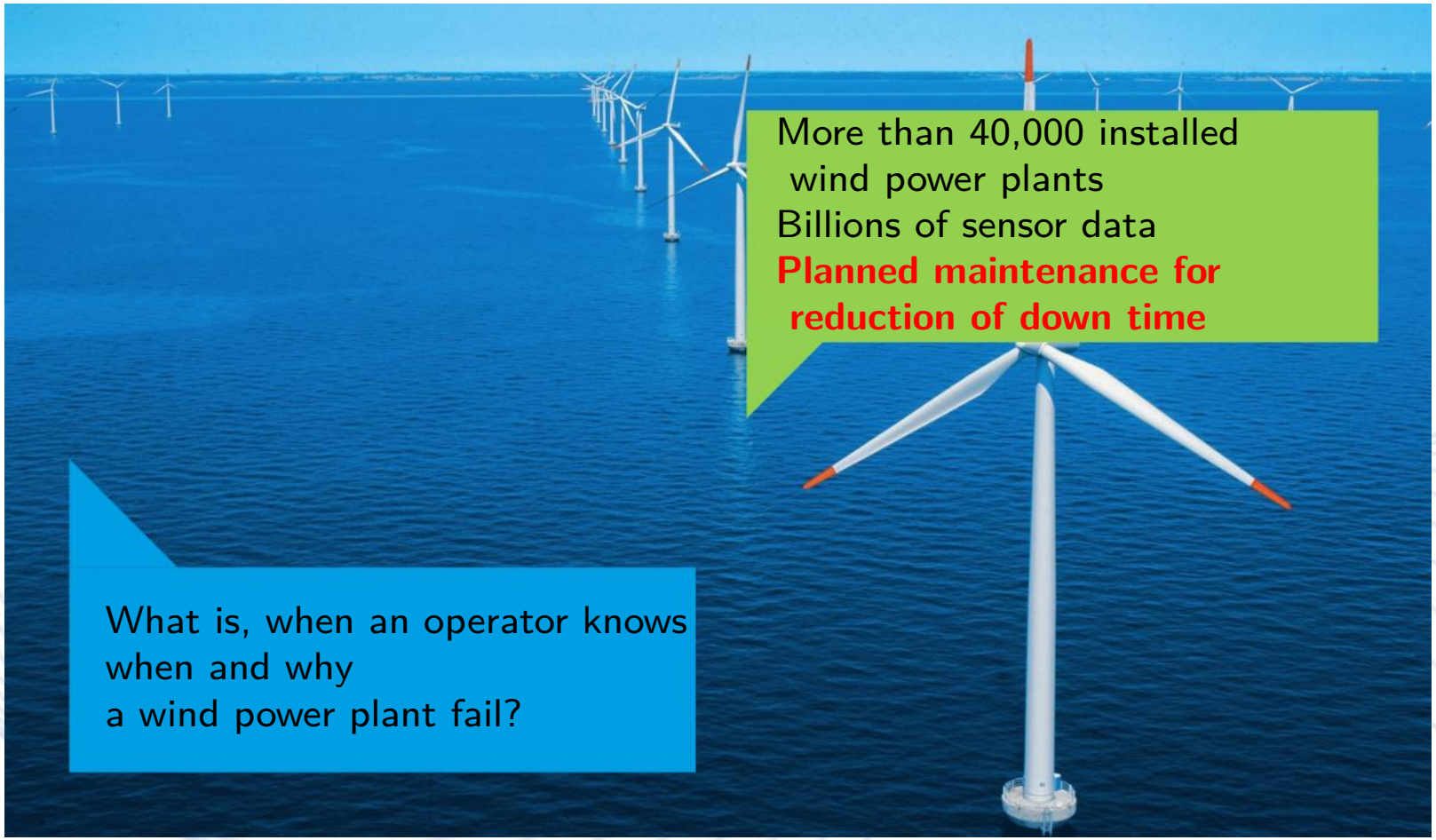

Example Energy

Example Energy

Example Energy

Example Fruits

Example Fruits

Example Fruits

The Mouse Game

or:

your will is predictive

https://github.com/SunOner/mouse_net

Facebook knows better as your friends or family

2015: Study with 86.220 volunteers

- Cooperation of the university Cambridge and Stanford
- Survey with 100 elements of the five-faktor-model (FFM) for the personality psychology / "Big Five"
  $\Rightarrow$ neuroticism, extraversion, openness for expiriences, conscience and tolerance
- Computer-Algorithm (linear regression) vs. assessment for persons
- with 10 Likes: Computer is better than collegues
- with 70 Likes: Computer is better than friends
- with 150 Likes: Computer is better than the family
- with 300 Likes: Computer is better than the partner

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⇒ A standard Facebook user has 227 Likes

⇒ Self-test: http://applymagicsauce.com/you.html

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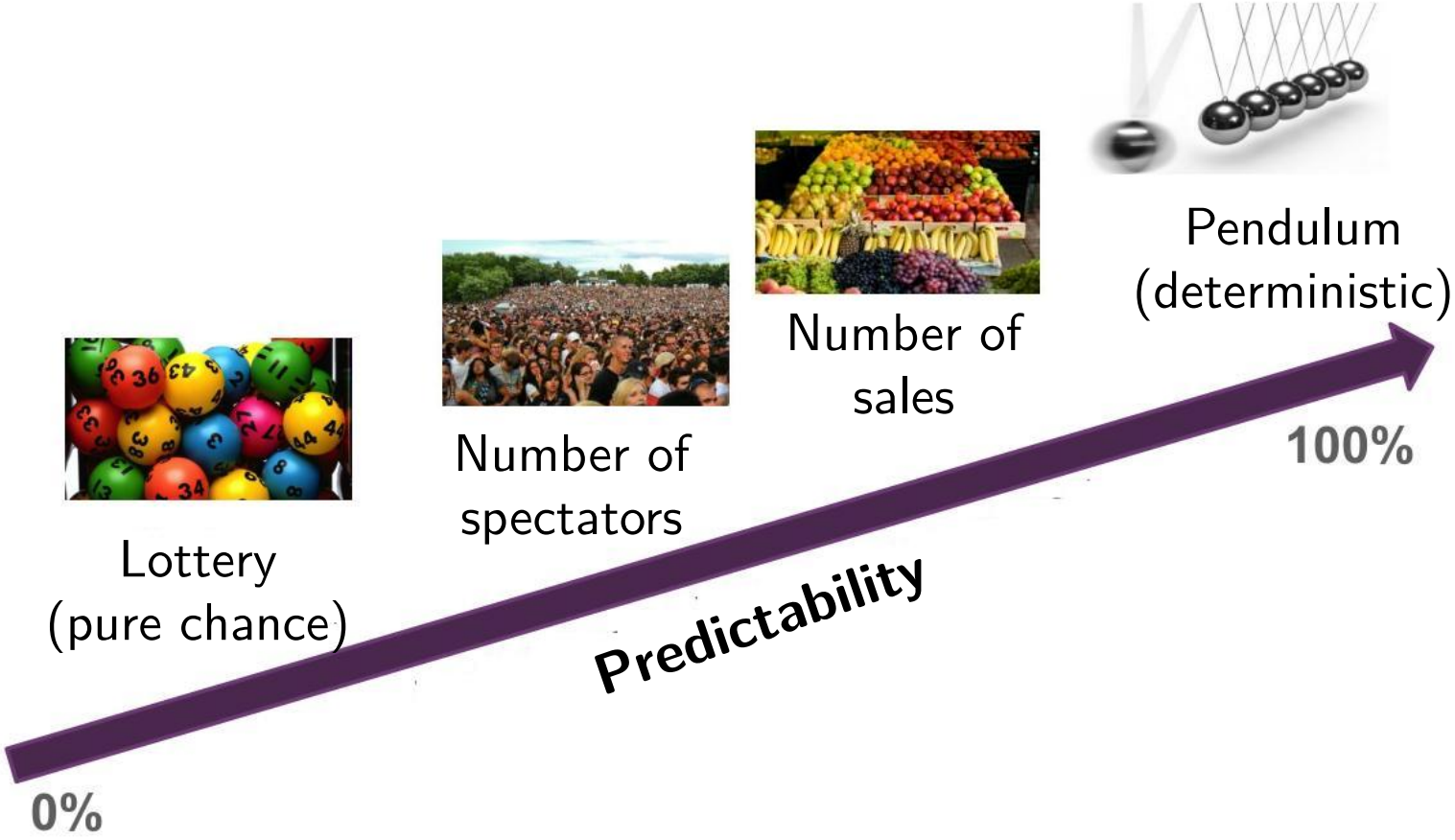
Amazon knows what you will buy

Predictive Analytics - Examples



Data Science

Complex BI Systems vs. Spreadsheet / Excel

- "90% of sheets with more than 150 rows has a least one error in a formula."
  Quelle: Freeman, D.: How to Make Spreadsheets Error-Proof, Journal of Accountancy 181 (1996) 75 - 77.
- "Significant Erros were in 20 of 22 sheets (91%). Nevertheless, 81% of user thinks that they have advantages comparing with their competitor using the sheets."
  Quelle: KPMG Management Consulting: Supporting the Decision Maker: A Guide to the Value of Business Modeling, 30.07.1998.
- "Seven idependent studies examined 113 sheets; 88% had big bugs."
  Quelle: Panko, R.R.: What We Know About Spreadsheet Errors, Journal of End User Computing's 10 (1998) 15 - 21

Data Scientist

- Data Scientist
  - The "Hot new gig in town" (O'Reilly report)
  - The Sexiest Job of the 21st Century
    - Harvard Business Review, October 2012
    - The next sexy job in next 10 years will be statistician - Hal Varian (Google Chief Economist)
    - Geek Chic - Wall Street Journal - new cool kids on campus
- The future belongs to the companies and people that turn data into products
- "The human expertise to capture and analyze big data is both the most expensive and the most constraining factor for most organizations pursuing big data initiatives "
  Thomas Davenport

Data Scientist

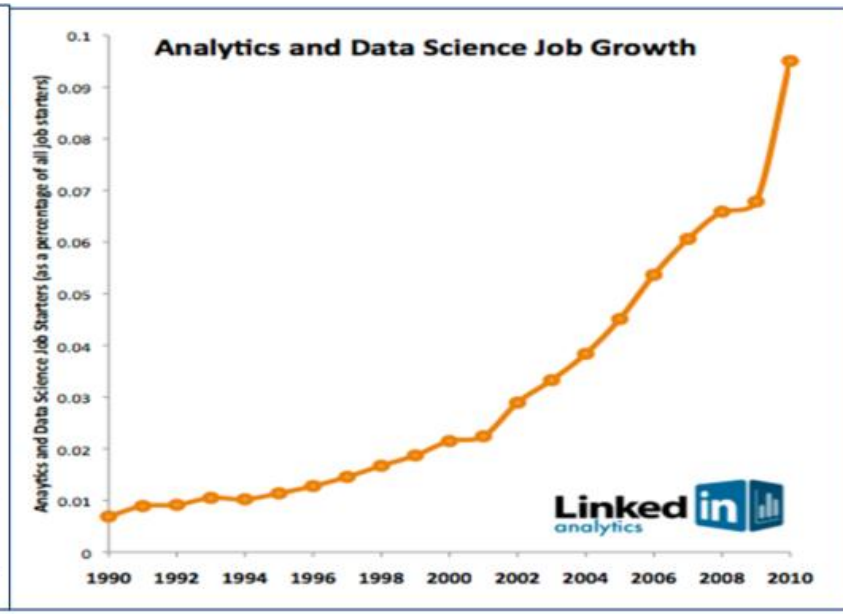
Article in Fortune

- "The unemployment rate in the U.S. continues to be abysmal (9.1% in 2016), but the tech world has spawned a new kind of highly skilled, nerdy-cool job that companies are scrambling to fill: data scientist"

McKinsey Global Institute "Big data Report"

- By 2018, the United States alone could face a shortage of 140,000 to 190,000 people with deep analytical skills as well as 1.5 million managers and analysts with the know-how to use the analysis of big data to make effective decisions

Data Science Job Growth



By 2018 shortage of 140-190,000 predictive analysts and 1,500,000 managers / analysts in the US

Data Miners: Past and Present

- Traditional approaches have been for DM experts: "White-coat PhD statisticians"

  $\Rightarrow$ DM tools also fairly expensive

- Today: approach is designed for those with some Database/Analytics skills
  - DM built into DB, easy to use GUI, Workflows
  - Many jobs available from Statistical analyst to Data Scientist!

- Data Science: The Art of mathematically sophisticated data engineers delivering insights from data into business decisions and systems